

University of Sri Jayewardenepura

Faculty of Computing



StudyVault – AI powered note organizer

Project proposal

CCS3102 | CSE3102 – Mobile Computing

B.comp (Hons.) in Computer Science

B.comp (Hons.) in Software Engineering

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Group 20 –

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Problem Statement

In this technological era, students do not rely on heavy old books to study. They often use digital learning materials such as PDF notes, lecture slides and images of handwritten content. However, managing these files efficiently in their personal devices could be tiresome. Many students rely on platforms such as WhatsApp, Google Drive and Local storage to store these files which often leads to disorganization and difficulty in finding any material.

This lack of a centralized and structured system to organize these learning materials is a key issue faced by many students. These files, being poorly named, scattered and duplicated makes it even harder to locate the needed files. This problem is a huge concern during exam periods as quick access to important content is critical.

Furthermore, Students spend countless hours reading through these PDF files without being able to instantly summarize or highlight key files. This can be quite a problem with the already limited time students possess, further reducing their productivity. Most existing applications focus on storage or note taking without any AI-based summarization and prioritization tools.

Therefore, Students are in need of an application which allows them to,

- Organize their notes
- Highlight/Pin key files
- Summarize learning materials

all in a centralized location.

Objectives

StudyVault mainly accomplishes the following objectives.

Primary Objectives

- Provide a Centralized platform to store and organize study materials (PDFs and images)
- Allow students to manage files efficiently
- Integrate AI-powered summarization for quick understanding

Secondary Objectives

- Allow students to mark important notes
- Enable secure user authentication
- Support file upload and download functionalities
- Improve student productivity by reducing time spent searching and reading learning materials

System Outline

StudyVault is a mobile application developed to help students manage their academic materials in a structured and intelligent way. This system allows students to upload files, organize them and use AI features to enhance their learning experience.

How the System works

1. User creates an account and securely logs in
2. User uploads PDF or image-based notes
3. Files are stored in a cloud database
4. Users can organize and manage files within the app
5. Users can generate summaries for selected files
6. Users can mark files as important

Functional requirements

- User registration and login
 - Students should be able to securely register and login to the system
- Upload PDF and image files
 - Students should be able to upload their study materials in the form of PDFs and image files.
- View and manage uploaded files
 - After uploading their learning materials, students should be able to view them and manage them within the platform.
- Mark important files
 - Any study materials deemed important by the student, should be able to highlight/pin as important within the platform.
- AI-based PDF summarization
 - Students should be able to use the AI functionality within the application to summarize their learning materials and save time.
- Basic file organization
 - The platform should provide the basic file organization capabilities to the user.

Non-functional requirements

- Simple and user-friendly User interface
 - Users should be able to navigate the UI seamlessly.
- Fast file upload
 - Students should be able to upload their learning materials to the application in a limited amount of time.
- Secure authentication and data storage
 - Student details and their learning materials should be stored within the database as securely as possible.
- Ability to handle multiple users and files

- The platform should be able to handle multiple users and files concurrently without a performance degradation.
- Stable performance without crashers
 - The platform should perform without any crashers.

Tools and Technologies

To develop StudyVault the following technologies will be used.

Frontend development

For Frontend development we will mainly be using the following,

- **Language:** Java
- **IDE:** Android Studio
- **UI Design:** XML layouts

Backend & Database

Frontend will be developed using Firebase or Supabase. It will be used for the following tasks,

- Authentication (User login/signup)
- Cloud storage (for files)
- Realtime database / Firestore (metadata)

AI integration

AI based summarization API will be used through Gemini API. It will be used to generate short summaries from uploaded PDFs. The users should be able to select the necessary file and call for the AI to summarize the files content.

Other tools

- GitHub - Version control
- Postman - API testing
- Figma - UI/UX design

Task Allocation

The development process will be carried out in a collaborative manner among the team members. They will be assigned roles with specific responsibilities and technical areas. However, each member will help each other in the necessary technical area to complete his/her work. Regular discussions and progress reviews will be conducted to ensure proper coordination among team members within the development. GitHub will be used for version control and task management throughout the development process.

The development tasks will be allocated as follows,

Name	Role	Tasks
T .G.Kithmi Pabodha	Frontend Developer	- Design UI layouts (login, dashboard, file view) - Implement file upload interface - Handle user interactions (buttons, navigation)
M. D. C. E. Gunathilaka	Backend Developer	- Set up Firebase/Supabase - Implement authentication system - Manage database structure and file storage
M. A. O. Rodrigo	AI Integration Developer	- Integrate AI summarization API - Process PDF files for text extraction - Display summaries in the app
All members	Testing and Documentation	- UI testing - Bug fixing - Presentation preparation - Report writing

GANTT chart

Process	May				June				July			
	1	2	3	4	1	2	3	4	1	2	3	4
Planning												
Development												
Testing												
Finalization												

Conclusion

StudyVault is a mobile application that addresses the common issue where students find it challenging to organize and manage their digital learning materials. With the increasing online resources used to learn, students depend on PDF notes and image-based study resources. Students are in need of a centralized platform that allows efficient access to their academic content.

The proposed system combines file management with AI-powered summarization features to improve productivity and simplify the learning process. With secure authentication, file uploading, note organization, highlighting important files and generating summaries, StudyVault aims to provide a convenient and user-friendly learning assistant for students.

The application will be developed using modern mobile development technologies such as Android studio, Java, Firebase/Supabase and Gemini AI integration. By implementing these technologies, the application is expected to deliver a reliable, secure and efficient user experience.

Overall, StudyVault aims to be a practical academic companion that helps students save time, stay organized and improve their overall learning experience through AI-powered digital note management.